

701. Insert into a Binary Search Tree

Solved

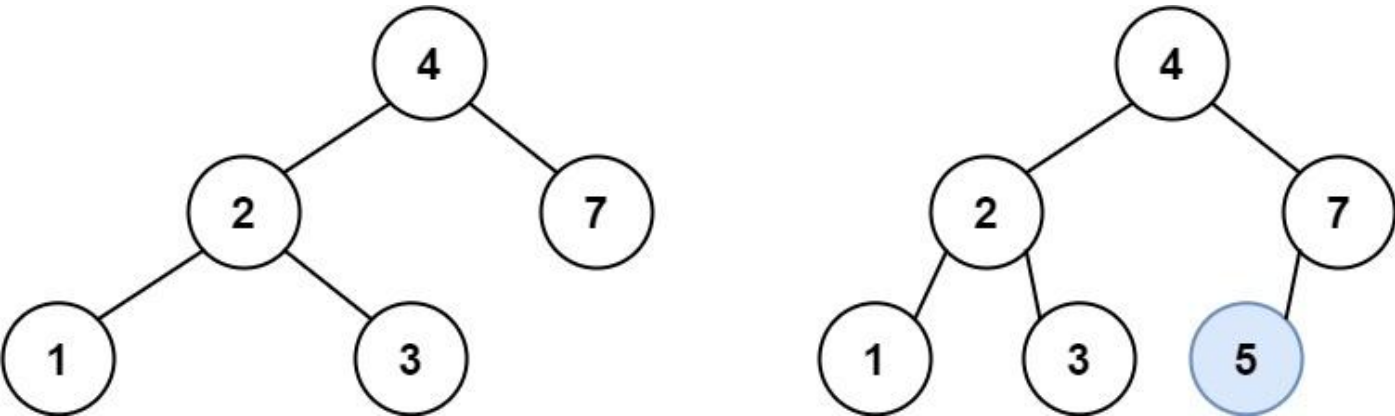
Medium

Topics

You are given the root node of a binary search tree (BST) and a value to insert into the tree. Return *the root node of the BST after the insertion*. It is **guaranteed** that the new value does not exist in the original BST.

Notice that there may exist multiple valid ways for the insertion, as long as the tree remains a BST after insertion. You can return **any of them**.

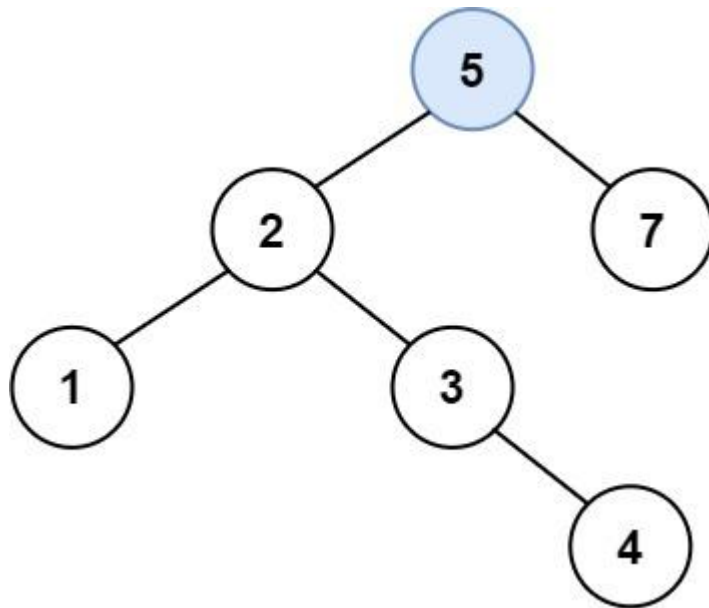
Example 1:



Input: root = [4,2,7,1,3], val = 5

Output: [4,2,7,1,3,5]

Explanation: Another accepted tree is:



Example 2:

Input: root = [40,20,60,10,30,50,70], val = 25

Output: [40,20,60,10,30,50,70,null,null,25]

Example 3:

Input: root = [4,2,7,1,3,null,null,null,null,null], val = 5

Output: [4,2,7,1,3,5]